

1. INTRODUCTION

As complexity of distribution network has grown, automation of substation has become a need of every utility company. To Improve the quality of power it is necessary to be familiar with what sort of constraint has occurred. Additionally, if there is any inadequacy in the protection, monitoring and control of a power system. Therefore, it necessitates a monitoring system that will be able to automatically detect, monitor, and classify the existing constraints on electrical lines.

Electricity is an extremely handy and useful form of energy. It plays an ever-growing role in our modern industrialized society. The electrical power systems are highly non-linear, extremely huge and complex networks. Such electric power systems are unified for economic benefits, increased reliability and operational advantages.

They are one of the most significant elements of both national and global infrastructure, and when these systems collapse it leads to major direct and indirect impacts on the economy and national security. A power system consists of components such as generators, lines, transformers, loads, switches and compensators.

However, a widely dispersed power sources and loads are the general configuration of modern power systems. Today electricity still suffers from power outages and blackouts due to the lack of automated analysis and poor visibility of the utility over the grid. WSN will give the utility provide the needed view by collecting information from the different sub-systems of the grid.

A sensor node will decide information or to slightly delay this notification (whether to immediately notify the sink about this information.). As complexity of distribution network has grown, automation of substation has become a need of every utility company to increase its efficiency and to improve quality of power being delivered.

1.1 Objectives

The main objectives of the project are:-

1. To improve reliability and compatibility.
2. Real time monitoring.
3. Remote sensing of observation parameters.

4. Remote sensing of observant parameters.
5. To Maintain Continuity of supply.
6. To Reduce labour cost.

1.2 Project Overview

The purpose of this project is to acquire the remote electrical parameters like voltage, current and frequency and send these real time values over network using IoT module along with temperature at power station. This project is also designed to protect the electrical circuitry by operating an SPDT relay. This relay gets activated whenever the electrical parameters exceed the predefined values. The relay can be used to switch off the main electrical supply. User can send commands to the microcontroller to read the remote electrical parameters.

This system also can automatically send the real time electrical parameters periodically (based on time settings). This system can be designed to send alerts whenever the relay trips or whenever the voltage or current exceeds the predefined limits. This project makes use of a microcontroller, in this prototype for demonstration purpose we have used Arduino Uno. The controller can efficiently communicate with the different sensors being used. The controller is provided with some internal memory to hold the code. This memory is used to dump some set of assembly instructions into the controller.

The project “**Substation Monitoring and Controlling based on Microcontroller by using GSM Modem**” was designed to allow devices to be monitored and controlled from anywhere in the world using mobile phone-connected GSM modem.

1.3 Organization of the report

The report explains the implementation of Substation Monitoring and Controlling based on Microcontroller by using GSM Modem using ATMEGA328P microcontroller. The organization of the thesis is explained here with:

Chapter 2: Deals with the substation design, construction, budgeting, site selection process.

Chapter 3: Deals with the required theory to understand what actually the embedded systems are and its need and applications.

Chapter 4: Deals with the hardware materials used in the project and their description.

Chapter 5: Deals with the software used in the project, its explanation and software compiling process.

Chapter 6: Deals with circuit diagram, test and results.

Chapter 7: Deals with all the future scopes related to the project and includes concluding statement of our work.

2.SUBSTATION

A substation is a part of an electrical generation, transmission and distribution system. Substations transform voltage from high to low, or the reverse, or perform any of several other important functions. Between the generating stations and consumer, electric power may flow through several substations at different voltage levels.

A substation may include transformers to change voltage levels between high transmission voltages and lower distribution voltages, or at the interconnection of two different transmission voltages. They are a common component of the infrastructure, for instance there are many number of substations across the country.

Substations may be owned and operated by an electrical utility, or may be owned by a large industrial or commercial customers. Generally substations are unattended, relying on SCADA for remote supervision and control. The word substation comes from the days before the distribution system became a grid.

As central generation stations became larger, smaller generating plants were converted to distribution stations, receiving their energy supply from a larger plant instead of their own generators, the first substations were connected to only one power station, where the generators were housed, and were subsidiaries of that power station.

2.1 Background

The construction of new substations and the expansion of existing facilities are commonplace projects in electric utilities. However, due to its complexity, very few utility employees are familiar with the complete process that allows these projects to be successfully completed. This chapter will attempt to highlight the major issues associated with these capital-intensive construction projects, and provide a basic understanding of the types of issues that must be addressed during this process.

There are four major types of electric substations. The first type is the switchyard at a generating station. These facilities connect the generators to the utility grid and also provide off-site power to the plant. Generator switchyards tend to be large installations that are typically engineered and constructed by the power-plant designers and are subject to planning, finance, and construction efforts different from those of routine substation projects. Because of their

special nature, the creation of power-plant switchyards will not be discussed here, but the expansion and modifications of these facilities generally follow the same processes as system stations.

Another type of substation, typically known as the customer substation, functions as the main source of electric power supply for one particular business customer. The technical requirements and the business case for this type of facility depend highly on the customer's requirements, more so than on utility needs; so this type of station will also not be the primary focus of this discussion.

The third type of substation involves the transfer of bulk power across the network, and is referred to as a system station. Some of these stations provide only switching facilities (no power transformers) whereas others perform voltage conversion as well. These large stations typically serve as the end points for transmission lines originating from generating switchyards and provide the electrical power for circuits that feed transformer stations.

They are integral to the long-term reliability and integrity of the electric system and enable large blocks of energy to be moved from the generators to the load centers. Since these system stations are strategic facilities and usually very expensive to construct and maintain, these substations will be one of the major focuses of this chapter.

The fourth type of substation is the distribution station. These are the most common facilities in power electric systems and provide the distribution circuits that directly supply most electric customers. They are typically located close to the load centers, meaning that they are usually located in or near the neighborhoods that they supply, and are the stations most likely to be encountered by the customers. Due to the large number of such substations, these facilities will also be a focus of this chapter.

Depending on the type of equipment used, the substations could be

- Outdoor type with air insulated equipment
- Indoor type with air insulated equipment
- Outdoor type with gas insulated equipment
- Indoor type with gas insulated equipment

- Mixed technology substations
- Mobile substations

2.2 Budgeting

Having established the broad requirements for the new station, such as voltages, capacity, number of feeders, etc., the issue of funding should then be addressed. This is typical when real-estate investigations of available sites begin, since site size and location can significantly affect the cost of the facility. Preliminary equipment layouts and engineering evaluations are also undertaken at this stage to develop ballpark costs, which then have to be evaluated in the corporate budgetary justification system.

Preliminary manpower forecasts of all disciplines involved in the engineering and construction of the substation should be undertaken, including identification of the nature and extent of any work that the utility may need to contract out. This budgeting process will involve evaluation of the project in light of corporate priorities and provide a general overview of cost and other resource requirements. Note that this process may be an annual occurrence. Any projects in which monies have yet to be spent are generally reevaluated every budget cycle.

2.3 Financing

Once the time has arrived for work to proceed on the project, the process of obtaining funding for the project must be started. Preliminary detailed designs are required to develop firm pricing. Coordination between business units is necessary to develop accurate costs and to develop a realistic schedule. This may involve detailed manpower forecasting in many areas. The resource information has to be compiled in the format necessary to be submitted to the corporate capital estimate system and internal presentations must be conducted to sell the project to all levels of management.

Sometimes it may be necessary to obtain funding to develop the capital estimate. This may be the case when the cost to develop the preliminary designs is beyond normal departmental budgets, or if unfamiliar technology is expected to be implemented. This can also occur on large, complex projects or when a major portion of the work will be contracted. It may also be necessary to obtain early partial funding in cases where expensive, long lead-time equipment may need to be purchased such as large power transformers.

2.4 Site Selection and Acquisition

At this stage, a footprint of the station has been developed, including the layout of the major equipment. A decision on the final location of the facility can now be made and various options can be evaluated.

As a rule of thumb, the following site evaluation criteria could be used:

- Economical evaluation
- Technical evaluation
- Community acceptance

Economical evaluation should address the level of affordability, return on investment, initial capital cost and life cycle cost. Technical aspects that can influence the site selection process could include the following:

- Land: choose areas that minimize the need for earth movement and soil disposal.
- Water: avoid interference with the natural drainage network.
- Vegetation: choose low productivity farming areas or uncultivated land.
- Protected areas: avoid any areas or spots listed as protected areas.
- Community planning: avoid urban areas, development land, or land held in reserve for future development.
- Community involvement: engage community in the approval process.
- Topography: flat but not prone to flood or water stagnation.
- Soil: suitable for construction of roads and foundations; low soil resistivity is desirable.
- Access: easy access to and from the site for transportation of large equipment, operators, and maintenance teams.
- Line entries: establishment of line corridors (alternatives: multi-circuit pylons, UG lines).
- Pollution: risk of equipment failure and maintenance costs increase with pollution level.

To address community acceptance issues it is recommended to

- Adopt a low profile layout with rigid buses supported on insulators over solid shape steel structures.
- Locate substations in visually screened areas (hills, forest), other buildings, and trees.
- Use gas insulated switchgear (GIS).
- Use colors, lighting.
- Use underground egresses as opposed to overhead.

2.5 Design, Construction, and Commissioning Process

Having selected the site location, the design construction and commissioning process would broadly follow the steps shown in fig 2.1. Recent trends in utilities have been toward sourcing design and construction of substations through competitive bidding process to ensure capital efficiency and labor productivity.

2.5.1 Station Design

The following can be used as a guide for various design elements:

Basic Layout

- Stage development diagram
- Bus configuration to meet single line requirements
- Location of major equipment and steel structures based on single line diagram
- General concept of station
- Electrical and safety clearances
- Ultimate stage

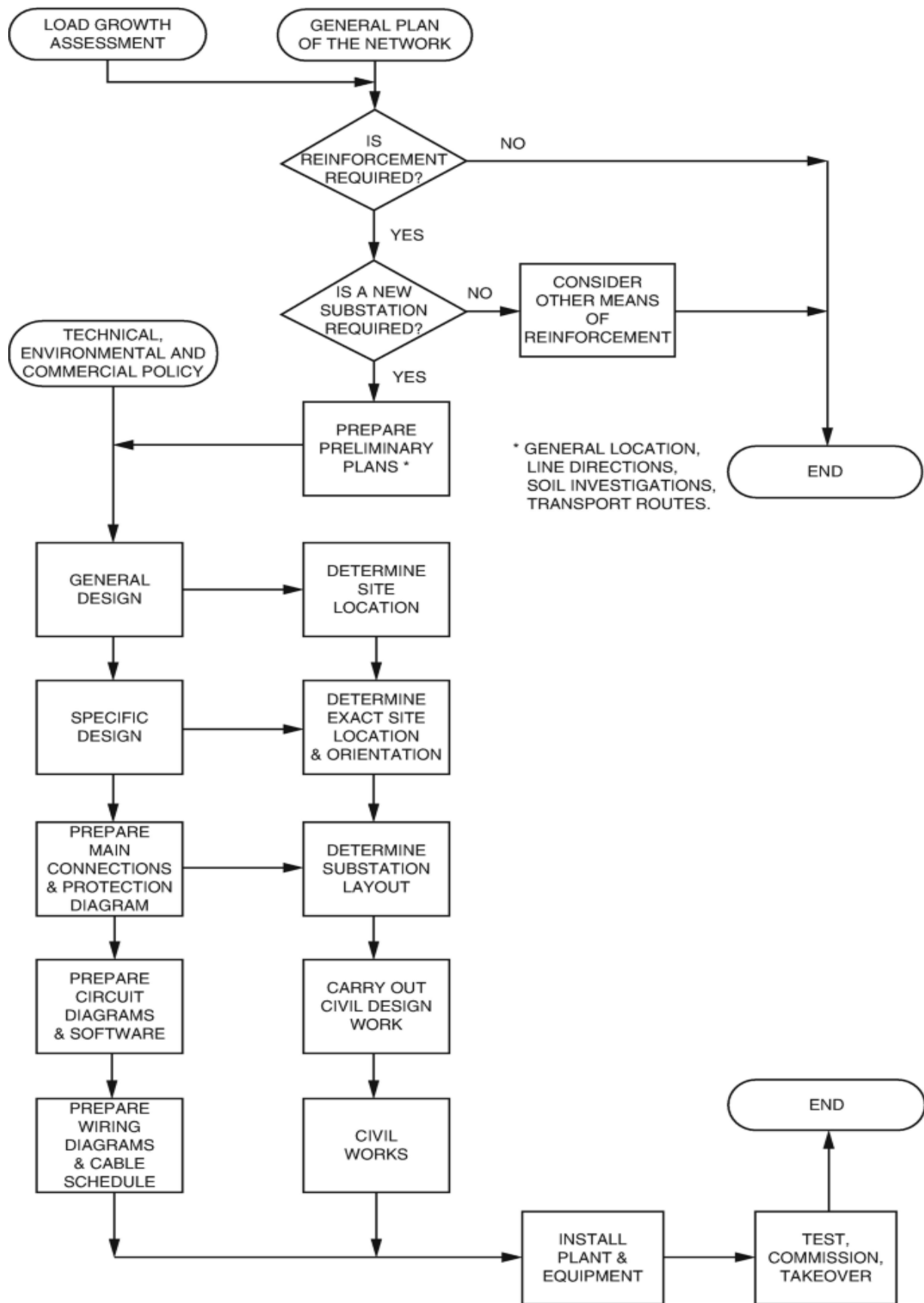


Fig. 2.1: Establishment of a new substation

2.5.2 Station Construction

With permits in hand and drawings published, the construction of the station can begin. Site logistics and housekeeping can have a significant impact on the acceptance of the facility. Parking for construction personnel, traffic routing, truck activity, trailers, fencing, and mud and dirt control along with trash and noise can be major irritations for neighbors, so attention to these details is essential for achieving community acceptance. All the civil, electrical, and electronic systems are installed at this time. Proper attention should also be paid to site security during the construction phase not only to safeguard the material and equipment, but also to protect the public.

2.5.3 Station Commissioning

Once construction is complete, testing of various systems can commence and all punch-list items addressed. To avoid duplication of testing, it is recommended to develop an inspection, testing and acceptance plan (ITAP). Elements of ITAP include

- Factory acceptance tests (FAT)
- Product verification plan (PVP)
- Site delivery acceptance test (SDAT)
- Site acceptance tests (SAT)

Final tests of the completed substation in a partially energized environment to determine acceptability and conformance to customer requirements under conditions as close as possible to normal operation conditions will finalize the in-service tests and turn-over to operations.

Environmental cleanup must be undertaken and final landscaping can be installed. Note that, depending upon the species of plants involved, it may be prudent to delay final landscaping until a more favorable season in order to ensure optimal survival of the foliage. Public relations personnel can make the residents and community leaders aware that the project is complete and the station can be made functional and turned over to the operating staff.

3.EMBEDDED SYSTEMS

3.1 Embedded Systems

An embedded system is a computer system designed to perform one or a few dedicated functions often with real-time computing constraints. It is embedded as part of a complete device often including hardware and mechanical parts. By contrast, a general-purpose computer, such as a personal computer (PC), is designed to be flexible and to meet a wide range of end-user needs. Embedded systems control many devices in common use today.

Embedded systems are controlled by one or more main processing cores that are typically either microcontrollers or digital signal processors (DSP). The key characteristic, however, is being dedicated to handle a particular task, which may require very powerful processors. For example, air traffic control systems may usefully be viewed as embedded, even though they involve mainframe computers and dedicated regional and national networks between airports and radar sites. (Each radar probably includes one or more embedded systems of its own.)

Since the embedded system is dedicated to specific tasks, design engineers can optimize it to reduce the size and cost of the product and increase the reliability and performance. Some embedded systems are mass-produced, benefiting from economies of scale.

Physically embedded systems range from portable devices such as digital watches and MP3 players, to large stationary installations like traffic lights, factory controllers, or the systems controlling nuclear power plants. Complexity varies from low, with a single microcontroller chip, to very high with multiple units, peripherals and networks mounted inside a large chassis or enclosure.

In general, "embedded system" is not a strictly definable term, as most systems have some element of extensibility or programmability. For example, handheld computers share some elements with embedded systems such as the operating systems and microprocessors which power them, but they allow different applications to be loaded and peripherals to be connected. Moreover, even systems which don't expose programmability as a primary feature generally need to support software updates. On a continuum from "general purpose" to "embedded", large application systems will have subcomponents at most points even if the system as a whole is "designed to perform one or a few dedicated functions", and is thus appropriate to call "embedded". A modern example of embedded system is shown in fig:3.1

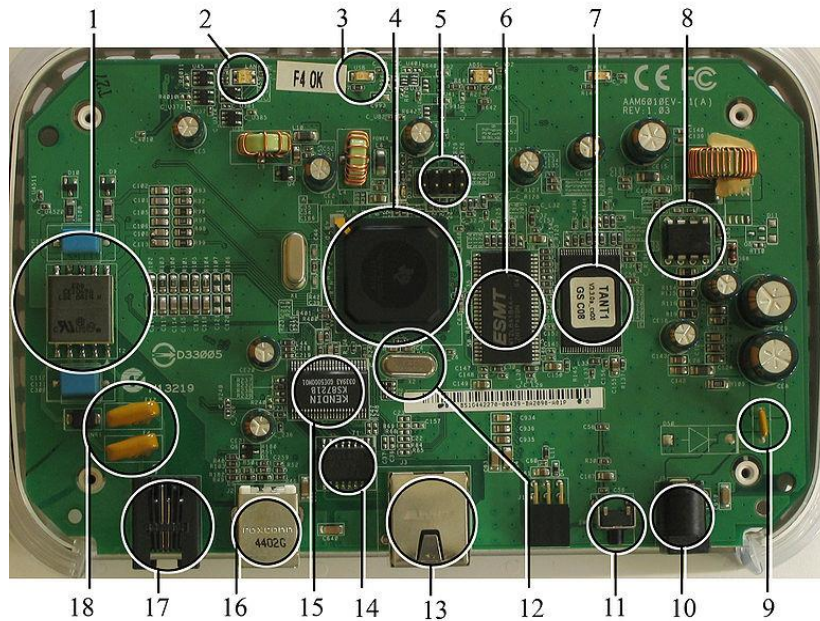


Fig 3.1: A modern example of embedded system

Labeled parts include microprocessor (4), RAM (6), flash memory (7). Embedded systems programming is not like normal PC programming. In many ways, programming for an embedded system is like programming PC 15 years ago. The hardware for the system is usually chosen to make the device as cheap as possible. Spending an extra dollar a unit in order to make things easier to program can cost millions. Hiring a programmer for an extra month is cheap in comparison. This means the programmer must make do with slow processors and low memory, while at the same time battling a need for efficiency not seen in most PC applications. Below is a list of issues specific to the embedded field.

3.1.1 History

In the earliest years of computers in the 1930–40s, computers were sometimes dedicated to a single task, but were far too large and expensive for most kinds of tasks performed by embedded computers of today. Over time however, the concept of programmable controllers evolved from traditional electromechanical sequencers, via solid state devices, to the use of computer technology.

One of the first recognizably modern embedded systems was the Apollo Guidance Computer, developed by Charles Stark Draper at the MIT Instrumentation Laboratory. At the project's inception, the Apollo guidance computer was considered the riskiest item in the Apollo project as it employed the then newly developed monolithic integrated circuits to reduce the size and weight. An early mass-produced embedded system was the Autonotic D-17 guidance

computer for the Minuteman missile, released in 1961. It was built from transistor logic and had a hard disk for main memory. When the Minuteman II went into production in 1966, the D-17 was replaced with a new computer that was the first high-volume use of integrated circuits.

3.1.2 Tools

Embedded development makes up a small fraction of total programming. There's also a large number of embedded architectures, unlike the PC world where 1 instruction set rules, and the Unix world where there's only 3 or 4 major ones. This means that the tools are more expensive. It also means that they're lower featured, and less developed. On a major embedded project, at some point you will almost always find a compiler bug of some sort.

Debugging tools are another issue. Since you can't always run general programs on your embedded processor, you can't always run a debugger on it. This makes fixing your program difficult. Special hardware such as JTAG ports can overcome this issue in part. However, if you stop on a breakpoint when your system is controlling real world hardware (such as a motor), permanent equipment damage can occur. As a result, people doing embedded programming quickly become masters at using serial IO channels and error message style debugging.

3.1.3 Resources

To save costs, embedded systems frequently have the cheapest processors that can do the job. This means your programs need to be written as efficiently as possible. When dealing with large data sets, issues like memory cache misses that never matter in PC programming can hurt you. Luckily, this won't happen too often- use reasonably efficient algorithms to start, and optimize only when necessary. Of course, normal profilers won't work well, due to the same reason debuggers don't work well.

Memory is also an issue. For the same cost savings reasons, embedded systems usually have the least memory they can get away with. That means their algorithms must be memory efficient (unlike in PC programs, you will frequently sacrifice processor time for memory, rather than the reverse). It also means you can't afford to leak memory. Embedded applications generally use deterministic memory techniques and avoid the default "new" and "malloc" functions, so that leaks can be found and eliminated more easily. Other resources programmers expect may not even exist. For example, most embedded processors do not have hardware

FPU (Floating-Point Processing Unit). These resources either need to be emulated in software, or avoided altogether.

3.1.4 Real Time Issues

Embedded systems frequently control hardware, and must be able to respond to them in real time. Failure to do so could cause inaccuracy in measurements, or even damage hardware such as motors. This is made even more difficult by the lack of resources available. Almost all embedded systems need to be able to prioritize some tasks over others, and to be able to put off/skip low priority tasks such as UI in favor of high priority tasks like hardware control.

3.2 Need for Embedded Systems

The uses of embedded systems are virtually limitless, because every day new products are introduced to the market that utilizes embedded computers in novel ways. In recent years, hardware such as microprocessors, microcontrollers, and FPGA chips have become much cheaper. So when implementing a new form of control, it's wiser to just buy the generic chip and write your own custom software for it. Producing a custom-made chip to handle a particular task or set of tasks costs far more time and money.

Many embedded computers even come with extensive libraries, so that "writing your own software" becomes a very trivial task indeed. From an implementation viewpoint, there is a major difference between a computer and an embedded system. Embedded systems are often required to provide Real-Time response. The main elements that make embedded systems unique are its reliability and ease in debugging.

3.2.1 Debugging

Embedded debugging may be performed at different levels, depending on the facilities available. From simplest to most sophisticated they can be roughly grouped into the following areas:

- Interactive resident debugging, using the simple shell provided by the embedded operating system (e.g. Forth and Basic)
- External debugging using logging or serial port output to trace operation using either a monitor in flash or using a debug server like the Remedy Debugger which even works for heterogeneous multicore systems.
- An in-circuit debugger (ICD), a hardware device that connects to the microprocessor via a JTAG or Nexus interface. This allows the operation of the microprocessor to be

controlled externally, but is typically restricted to specific debugging capabilities in the processor.

- An in-circuit emulator replaces the microprocessor with a simulated equivalent, providing full control over all aspects of the microprocessor.
- A complete emulator provides a simulation of all aspects of the hardware, allowing all of it to be controlled and modified and allowing debugging on a normal PC.
- Unless restricted to external debugging, the programmer can typically load and run software through the tools, view the code running in the processor, and start or stop its operation. The view of the code may be as assembly code or source-code.

Because an embedded system is often composed of a wide variety of elements, the debugging strategy may vary. For instance, debugging a software(and microprocessor) centric embedded system is different from debugging an embedded system where most of the processing is performed by peripherals (DSP, FPGA, co-processor). An increasing number of embedded systems today use more than one single processor core. A common problem with multi-core development is the proper synchronization of software execution. In such a case, the embedded system design may wish to check the data traffic on the busses between the processor cores, which requires very low-level debugging, at signal/bus level, with a logic analyzer, for instance.

3.2.2 Reliability

Embedded systems often reside in machines that are expected to run continuously for years without errors and in some cases recover by themselves if an error occurs. Therefore the software is usually developed and tested more carefully than that for personal computers, and unreliable mechanical moving parts such as disk drives, switches or buttons are avoided.

Specific reliability issues may include:

- The system cannot safely be shut down for repair, or it is too inaccessible to repair. Examples include space systems, undersea cables, navigational beacons, bore-hole systems, and automobiles.
- The system must be kept running for safety reasons. "Limp modes" are less tolerable. Often backups are selected by an operator. Examples include aircraft navigation, reactor control systems, safety-critical chemical factory controls, train signals, engines on single-engine aircraft.

- The system will lose large amounts of money when shut down: Telephone switches, factory controls, bridge and elevator controls, funds transfer and market making, automated sales and service.

A variety of techniques are used, sometimes in combination, to recover from errors—both software bugs such as memory leaks, and also soft errors in the hardware:

- Watchdog timer that resets the computer unless the software periodically notifies the watchdog
- Subsystems with redundant spares that can be switched over to
- software "limp modes" that provide partial function
- Designing with a Trusted Computing Base (TCB) architecture[6] ensures a highly secure & reliable system environment
- An Embedded Hypervisor is able to provide secure encapsulation for any subsystem component, so that a compromised software component cannot interfere with other subsystems, or privileged-level system software. This encapsulation keeps faults from propagating from one subsystem to another, improving reliability. This may also allow a subsystem to be automatically shut down and restarted on fault detection.
- Immunity Aware Programming

3.3 Explanation of Embedded Systems

3.3.1 Software Architecture

There are several different types of software architecture in common use.

- Simple Control Loop:

In this design, the software simply has a loop. The loop calls subroutines, each of which manages a part of the hardware or software.

- Interrupt Controlled System:

Some embedded systems are predominantly interrupt controlled. This means that tasks performed by the system are triggered by different kinds of events. An interrupt could be generated for example by a timer in a predefined frequency, or by a serial port controller receiving a byte. These kinds of systems are used if event handlers need low latency and the event handlers are short and simple.

Usually these kinds of systems run a simple task in a main loop also, but this task is not very sensitive to unexpected delays. Sometimes the interrupt handler will add longer tasks to a queue structure. Later, after the interrupt handler has finished, these tasks are executed by the main loop. This method brings the system close to a multitasking kernel with discrete processes.

- Cooperative Multitasking:

A non-preemptive multitasking system is very similar to the simple control loop scheme, except that the loop is hidden in an API. The programmer defines a series of tasks, and each task gets its own environment to “run” in. When a task is idle, it calls an idle routine, usually called “pause”, “wait”, “yield”, “nop” (stands for no operation), etc. The advantages and disadvantages are very similar to the control loop, except that adding new software is easier, by simply writing a new task, or adding to the queue-interpreter.

- Primitive Multitasking:

In this type of system, a low-level piece of code switches between tasks or threads based on a timer (connected to an interrupt). This is the level at which the system is generally considered to have an "operating system" kernel. Depending on how much functionality is required, it introduces more or less of the complexities of managing multiple tasks running conceptually in parallel.

As any code can potentially damage the data of another task (except in larger systems using an MMU) programs must be carefully designed and tested, and access to shared data must be controlled by some synchronization strategy, such as message queues, semaphores or a non-blocking synchronization scheme.

Because of these complexities, it is common for organizations to buy a real-time operating system, allowing the application programmers to concentrate on device functionality rather than operating system services, at least for large systems; smaller systems often cannot afford the overhead associated with a generic real time system, due to limitations regarding memory size, performance, and/or battery life.

- Microkernels And Exokernels:

A microkernel is a logical step up from a real-time OS. The usual arrangement is that the operating system kernel allocates memory and switches the CPU to different threads of

execution. User mode processes implement major functions such as file systems, network interfaces, etc.

In general, microkernels succeed when the task switching and intertask communication is fast, and fail when they are slow. Exokernels communicate efficiently by normal subroutine calls. The hardware and all the software in the system are available to, and extensible by application programmers. Based on performance, functionality, requirement the embedded systems are divided into three categories:

3.3.2 Stand Alone Embedded System

These systems takes the input in the form of electrical signals from transducers or commands from human beings such as pressing of a button etc., process them and produces desired output. This entire process of taking input, processing it and giving output is done in standalone mode. Such embedded systems comes understand alone embedded systems

e.g: microwave oven, air conditioner etc..

3.3.3 Real-Time Embedded Systems

Embedded systems which are used to perform a specific task or operation in a specific time period those systems are called as real-time embedded systems. There are two types of real-time embedded systems.

- Hard Real-time embedded systems:

These embedded systems follow an absolute dead line time period i.e., if the tasking is not done in a particular time period then there is a cause of damage to the entire equipment.

Eg: consider a system in which we have to open a valve within 30 milliseconds. If this valve is not opened in 30ms this may cause damage to the entire equipment. So in such cases we use embedded systems for doing automatic operations.

- Soft Real Time embedded systems:

These embedded systems follow a relative dead line time period i.e., if the task is not done in a particular time that will not cause damage to the equipment.

Eg: Consider a TV remote control system, if the remote control takes a few milliseconds delay it will not cause damage either to the TV or to the remote control. These systems which will

not cause damage when they are not operated at considerable time period those systems comes under soft real-time embedded systems.

3.3.4 Network Communication Embedded Systems

A wide range network interfacing communication is provided by using embedded systems.

- Consider a web camera that is connected to the computer with internet can be used to spread communication like sending pictures, images, videos etc., to another computer with internet connection throughout anywhere in the world.
- Consider a web camera that is connected at the door lock.

Whenever a person comes near the door, it captures the image of a person and sends to the desktop of your computer which is connected to internet. This gives an alerting message with image on to the desktop of your computer, and then you can open the door lock just by clicking the mouse. Fig: 2.2 show the network communications in embedded systems.



Fig. 3.2: Network communication embedded systems

3.3.5 Different Types of Processing Units

The central processing unit (CPU) can be any one of the following microprocessor, microcontroller, digital signal processing.

- Among these Microcontroller is of low cost processor and one of the main advantage of microcontrollers is, the components such as memory, serial communication

interfaces, analog to digital converters etc., all these are built on a single chip. The numbers of external components that are connected to it are very less according to the application.

- Microprocessors are more powerful than microcontrollers. They are used in major applications with a number of tasking requirements. But the microprocessor requires many external components like memory, serial communication, hard disk, input output ports etc., so the power consumption is also very high when compared to microcontrollers.
- Digital signal processing is used mainly for the applications that particularly involved with processing of signals

3.4 APPLICATIONS OF EMBEDDED SYSTEMS

3.4.1 Consumer Applications

At home we use a number of embedded systems which include microwave oven, remote control, VCD players, DVD players, camera etc....



Fig. 3.3: Automatic coffee makes equipment

3.4.2 Office Automation

We use systems like fax machine, modem, printer etc...



Fig. 3.4: Fax machine and Printing machine

3.4.3. Industrial Automation

Today a lot of industries are using embedded systems for process control. In industries we design the embedded systems to perform a specific operation like monitoring temperature, pressure, humidity, voltage, current etc., and basing on these monitored levels we do control other devices, we can send information to a centralized monitoring station.



Fig. 3.5: Robot

In critical industries where human presence is avoided there we can use robots which are programmed to do a specific operation.

3.4.4 Computer Networking

Embedded systems are used as bridges routers etc..



Fig. 3.6: Computer networking

3.4.5 Tele Communications

Cell phones, web cameras etc.



Fig. 3.7: Cell Phone and Web camera

4. HARDWARE COMPONENTS

The hardware components required for the project are:

1. Microcontroller
2. Arduino
3. Current Transformer
4. GSM Modem
5. Rectifier
6. Relay
7. Crystal Oscillator
8. LCD Display
9. Temperature Sensor

4.1 Microcontroller

A computer-on-a-chip is a variation of a microprocessor which combines the processor core (CPU), some memory, and I/O (input/output) lines, all on one chip. The computer-on-a-chip is called the microcomputer whose proper meaning is a computer using a (number of) microprocessor(s) as its CPUs, while the concept of the microcomputer is known to be a microcontroller. A microcontroller can be viewed as a set of digital logic circuits integrated on a single silicon chip. This chip is used for only specific applications.



Fig. 4.1: Microcontroller

A microcontroller can be considered a self-contained with a processor, memory and peripherals and can be used an embedded system. Microcontrollers are embedded inside devices to control the actions and features of a product. Hence they can also be referred to as embedded controllers. They run one specific program and dedicated to a single task. They are low power devices and small LED or LCD display outputs.

4.2 Arduino

Arduino is an open-source platform used for building electronics projects. Arduino consists of both a physical programmable circuit board and a piece of software, used to write and upload computer code to the physical board. Arduino boards are available at low cost. It uses a simplified programming syntax based on C/C++ languages.

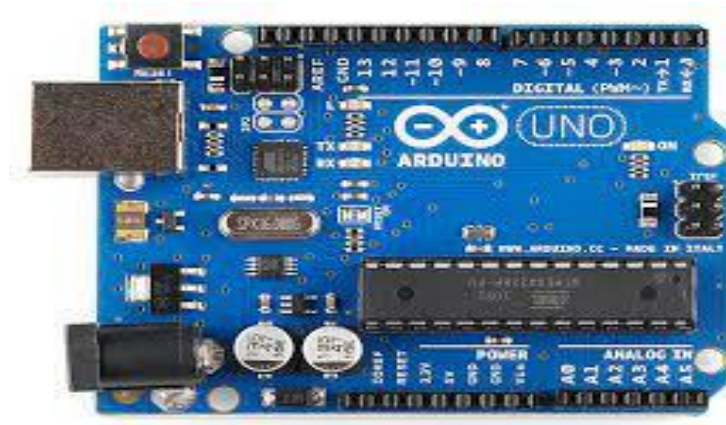


Fig. 4.2: Arduino UNO Board

This board contains a USB interface i.e.USB cable is used to connect the board with the computer and Arduino IDE (Integrated Development Environment) software is used to program the board.

4.3 Current Transformer

A current transformer (CT) is a type of transformer that is used to reduce or multiply an alternating current (AC). It produces a current in its secondary which is proportional to the current in its primary. Current transformers are used extensively for measuring current and monitoring the operation of the power grid. The alternating current in the primary produces an alternating magnetic field in the core, which then induces an alternating current in the secondary.

The primary circuit is largely unaffected by the insertion of the CT. Accurate current transformers need close coupling between the primary and secondary to ensure that the secondary current is proportional to the primary current over a wide current range. The current in the secondary is the current in the primary (assuming a single turn primary) divided by the number of turns of the secondary.



Fig. 4.3: Current Transformer

The current transformer reduces a high current to a lower value and therefore provides a safe way of monitoring electrical current flowing in an AC transmission line. Under normal operating conditions the secondary winding of a CT is connected to its burden, and it is always closed. When the current flows through the primary windings, it always flows through secondary windings and amperes turns of each winding are subsequently equal and opposite.

4.4 GSM Modem

GSM stands for Global System for Mobile Communication. The GSM Modem is used to send the real time parameters in the form of SMS. The system is designed to send SMS alerts to the authorized person whenever the parameters (Voltage, Current, frequency and Temperature) exceed the predefined limits. The GSM provides basic to advanced voice and data services including Roaming service. A GSM digitizes and compresses data, then sends it down through a channel. The sim used in the GSM modem is SIM800 having input voltage of 12V and the output obtained.



Fig. 4.4: GSM Modem

The Arduino GSM allows an Arduino board to connect to the internet, send and receive SMS, and make voice calls using the GSM library. The shield will work with the Arduino Uno out of the box.

4.5 Rectifier

A rectifier is an electrical device that converts alternating current to direct current DC, a process known as rectification. Rectifiers have many uses including as components of power supplies and as detectors of radio signals. Rectifiers may be made of solid-state diodes, vacuum tube diodes, mercury arc valves, and other components. When only one diode is used to rectify AC (by blocking the negative or positive portion of the waveform), the difference between the term diode and the term rectifier is merely one of usage, i.e., the term rectifier describes a diode that is being used to convert AC to DC. Almost all rectifiers comprise a number of diodes in a specific arrangement for more efficiently converting AC to DC than is possible with only one diode.



Fig. 4.5: Rectifiers

4.5.1 Bridge full wave rectifier

The Bridge rectifier circuit is shown in fig: 3.2, which converts an ac voltage to dc voltage using both half cycles of the input ac voltage. The Bridge rectifier circuit is shown in the figure. The circuit has four diodes connected to form a bridge. The ac input voltage is applied to the diagonally opposite ends of the bridge. The load resistance is connected between the other two ends of the bridge.

For the positive half cycle of the input ac voltage, diodes D1 and D3 conduct, whereas diodes D2 and D4 remain in the OFF state. The conducting diodes will be in series with the load resistance R_L and hence the load current flows through R_L .

For the negative half cycle of the input ac voltage, diodes D2 and D4 conduct whereas, D1 and D3 remain OFF. The conducting diodes D2 and D4 will be in series with the load resistance R_L and hence the current flows through R_L in the same direction as in the previous half cycle. Thus a bi-directional wave is converted into a unidirectional wave.

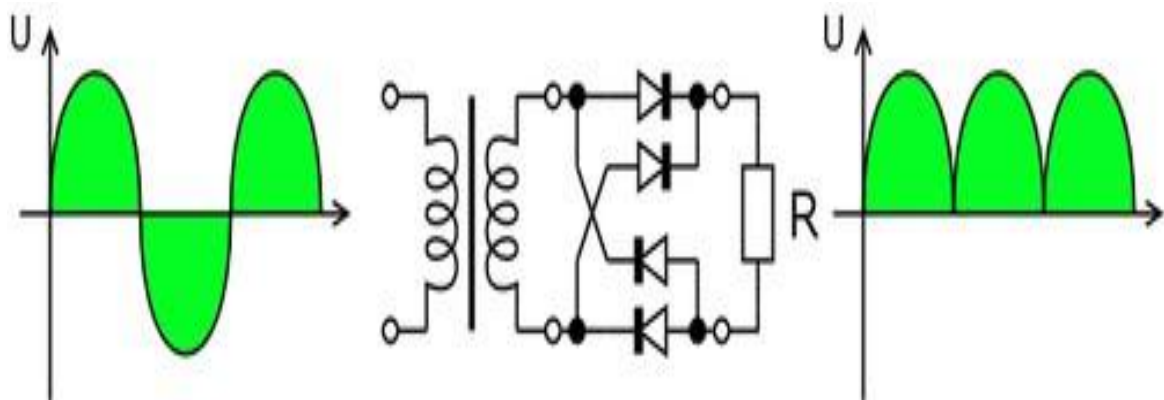


Fig. 4.6: Bridge rectifier- a full-wave rectifier using 4 diodes

4.6 Relay

Relay is an electromagnetic switch operated by a relatively small electric current that can turn on or off a much larger electric current. It consists of a set of input terminals for a single or multiple control signals, and a set of operating contact terminals. Because of unique characteristics, it is widely used in many fields.



Fig. 4.7: Relay

A magnetic field generated by the input signal operates on the mechanical contacts. When the current or voltage exceeds the rated value, then the relay operates and sends the information to circuit breaker then the circuit breaker makes or breaks the circuit. Relays are used where it is necessary to control a circuit by an independent low-power signal, or where several circuits must be controlled by one signal. Relays are not limited transforming single inputs to single outputs at single points in the circuit. a single relay can activate multiple circuits, allowing one input to initiate many other effects.

4.7 Crystal Oscillator

Crystal oscillator works on the principle of the Inverse Piezoelectric effect, the applied electrical field will generate a mechanical distortion across some material. Therefore, it utilizes the vibrating crystal's mechanical resonance, which is made through a piezoelectric material for generating an electrical signal of a certain frequency.

A crystal oscillator is an electronic oscillator circuit that uses the mechanical resonance of a vibrating crystal of piezoelectric material to create an electrical signal with a precise frequency. This frequency is often used to keep track of time, as in quartz wristwatches, to provide a stable clock signal for digital integrated circuits, and to stabilize frequencies for radio transmitters and receivers.

A crystal oscillator, particularly one made of quartz crystal, works by being distorted by an electric field when voltage is applied to an electrode near or on the crystal. This property is known as electrostriction or inverse piezoelectricity. When the field is removed, the quartz – which oscillates in a precise frequency – generates an electric field as it returns to its previous shape, and this can generate a voltage. The result is that a quartz crystal behaves like an RLC circuit.



Fig. 4.8: Crystal Oscillator

Quartz crystals are manufactured for frequencies from a few tens of kilohertz to hundreds of megahertz. More than two billion crystals are manufactured annually. Most are used for consumer devices such as wristwatches, clocks, radios, computers, and cellphones. Quartz crystals are also found inside test and measurement equipment, such as counters, signal generators, and oscilloscopes.

4.8 LCD Display

Nowadays, we always use devices which are made up of LCDs such as CD players, DVD players, digital watches, computers, etc. These are commonly used in the screen industries to replace the utilization of CRTs. Cathode Ray Tubes use huge power when compared with LCDs, and CRTs heavier as well as bigger. These devices are thinner as well and power consumption is extremely less. The LCD16x2 working principle is, it blocks the light rather than dissipates.

One of the most common devices attached to a microcontroller is an LCD display. Some of the most common LCD's connected to the many microcontrollers are 16x2 and 20x2 displays. This means 16 characters per line by 2 lines and 20 characters per line by 2 lines, respectively.

The term LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations, etc.



Fig. 4.9: LCD Display

The LCD requires 3 control lines as well as either 4 or 8 I/O lines for the data bus. The user may select whether the LCD is to operate with a 4-bit data bus or an 8-bit data bus. If a 4-bit data bus is used the LCD will require a total of 7 data lines (3 control lines plus the 4 lines for the data bus). If an 8-bit data bus is used the LCD will require a total of 11 data lines (3 control lines plus the 8 lines for the data bus).

4.9 Temperature Sensor

The temperature sensor LM35 sensor series are precision integrated-circuit temperature sensors, whose output voltage is linearly proportional to the Celsius temperature. The temperature sensor is used to detect the heat produced during fire occurrence. The LM35 sensor thus has an advantage over linear temperature sensors calibrated in degree kelvin, as the user is not required to subtract a large constant voltage from its output to obtain convenient Centigrade scaling.

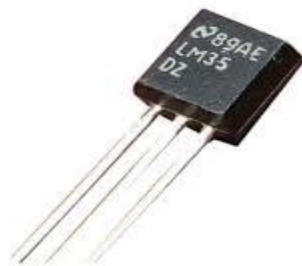


Fig. 4.10: Temperature Sensor

Thermistors, RTD's, Thermocouples, Temperature probes are different types of temperature sensors. Temperature sensors are precise and have a low cost. These are suitable for both embedded and surface applications. They provide low thermal mass resulting in fast response.

5. SOFTWARE REQUIREMENTS

5.1 USB Programmer

The USB Chip Programmer is, a flexible and simple to use programming device supporting a broad range of chips, including Eproms, EEproms and other storage chips. With a height of only 2.5cm, the enclosure is very practical and is noted for the design and anodized die-cast aluminum construction.

Serial transmission and reception between host and serial devices. The data transfer is of λ Controlled data transfer, Bulk data transfer, Interrupt driven data transfer, Iso-synchronous transfer. A bus between the host system and interconnected number of peripheral devices.

5.1.1 Working of USB

When a peripheral device is attached via USB, the host computer will detect what kind of device it is and automatically load a driver that allows the device to function. Data is transferred between the two devices in small amounts known as 'packets'. A connector is used for the upstream connection towards the host and B connector for the downward stream to the USB device. When the device is connected to the PC, it activates the host to recognize it. The PC detects the device and manages a control flow between the device and computer.

5.2 USB a Wiring Connection

The USB cable has typically four wires to connect the A type connector. PC also manages the data transfer between the device and PC. Once detected, the PC sends data to the USB system software to recognize it which then identify the device and assign an address. This address is used to detect the particular USB device. The software controls the input and output data between the PC and device. If the software fails to assign the address, PC will not detect the device.

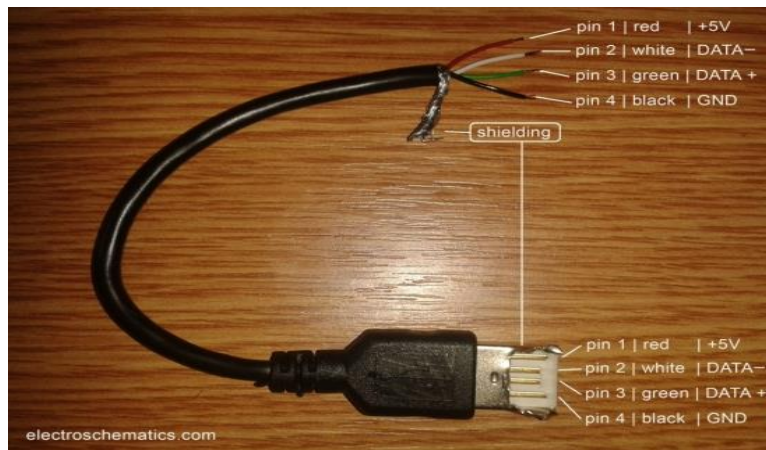


Fig. 5.1: USB connecting wire

5.3 USB Hub

The USB Hub is used to connect many devices to the PC using a single USB connector. The hub can detect the attachment or detachment of devices in each port of the Hub. It also distributes power to all the devices connected to it and also detects low speed and full speed devices.



Fig. 5.2: USB Hub Controller

It has two components – A Hub controller and a Hub repeater. The controller enables the Hub to communicate with the PC for configuration and control of devices attached to it. The repeater has hardware support for reset, suspend and resume signals.

I recommend you to use a hub with an external power supply if you want to connect more than 2 or 3 devices because they will absorb a large amount of current and the USB port cannot provide it by itself.

5.4 USB Flash Drive

The most commonly used USB device is the Flash drive the commonly called Pen drive. It is a mass storage device capable of functioning like a hard disk of computer.

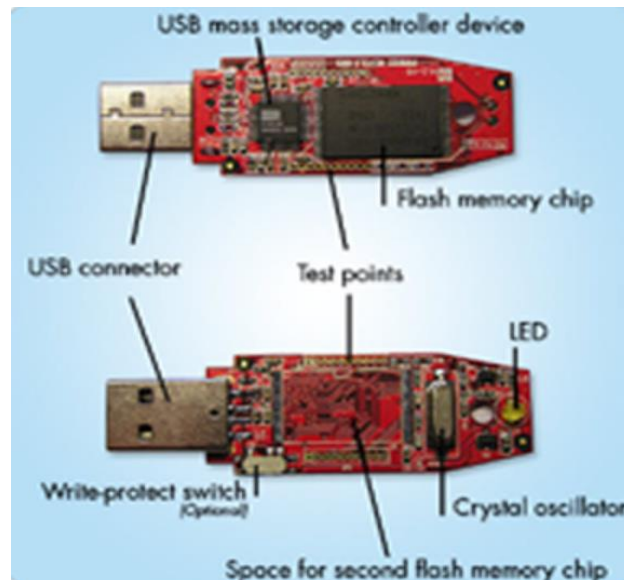


Fig 5.3: Typical USB Flash Drive

5.5 USB Device Features

- Can be hot plugged (attached), configured and used, reset, reconfigured and used Bandwidth sharing with other devices.
- Host schedules the sharing of bandwidth among the attached devices at an instance.
- Can be detached (while others are in operation) and reattached.
- Attaching and detaching USB device or host without rebooting.
- Can be hot plugged (attached), configured and used, reset, reconfigured and used.
- Bandwidth sharing with other devices.
- Host schedules the sharing of bandwidth among the attached devices at an instance.
- Can be detached (while others are in operation) and reattached.
- Attaching and detaching USB device or host without rebooting.

5.6 Proteus

Proteus is software which accepts only hex files. Once the machine code is converted into hex code, that hex code has to be dumped into the microcontroller and this is done by the Proteus. Proteus is a programmer which itself contains a microcontroller in it other than the one which is to be programmed. This microcontroller has a program in it written in such a way that it accepts the hex file from the pic compiler and dumps this hex file into the microcontroller which is to be programmed. As the Proteus programmer requires power supply to be operated, this power supply is given from the power supply circuit designed and connected to the microcontroller in proteus.

The program which is to be dumped into the microcontroller is edited in proteus and is compiled and executed to check any errors and hence after the successful compilation of the program the program is dumped into the microcontroller using a dumper.

4.4 Procedural steps for compilation, simulation and dumping:

4.4.1 Compilation and simulation steps: For the PIC microcontroller, the PIC C compiler is used for compilation. The compilation steps are as follows: 52

- Open PIC C compiler.
- You will be prompted to choose a name for the new project, so create a separate folder where all the files of your project will be stored, choose a name and click save.

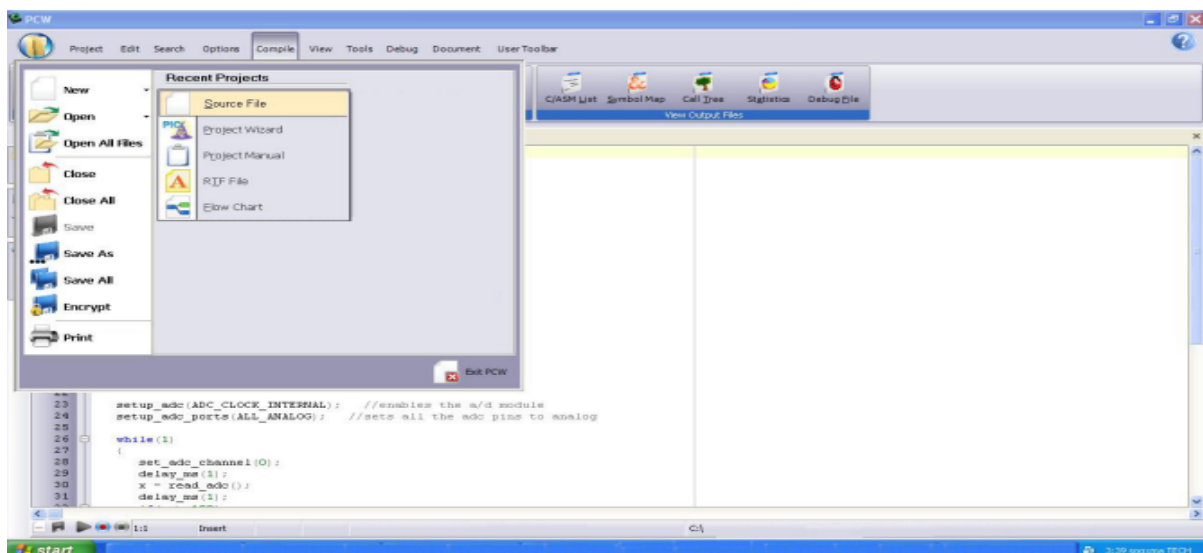


Fig. 5.4: Picture of opening a new file using PIC C compiler

- Click Project, New, and something the box named 'Text1' is where your code should be written later.

- Now you have to click ‘File, Save as’ and choose a file name for your source code ending with the letter ‘.c’. We can name it as ‘project.c’ for example and click save. Then you have to add this file to your project work.

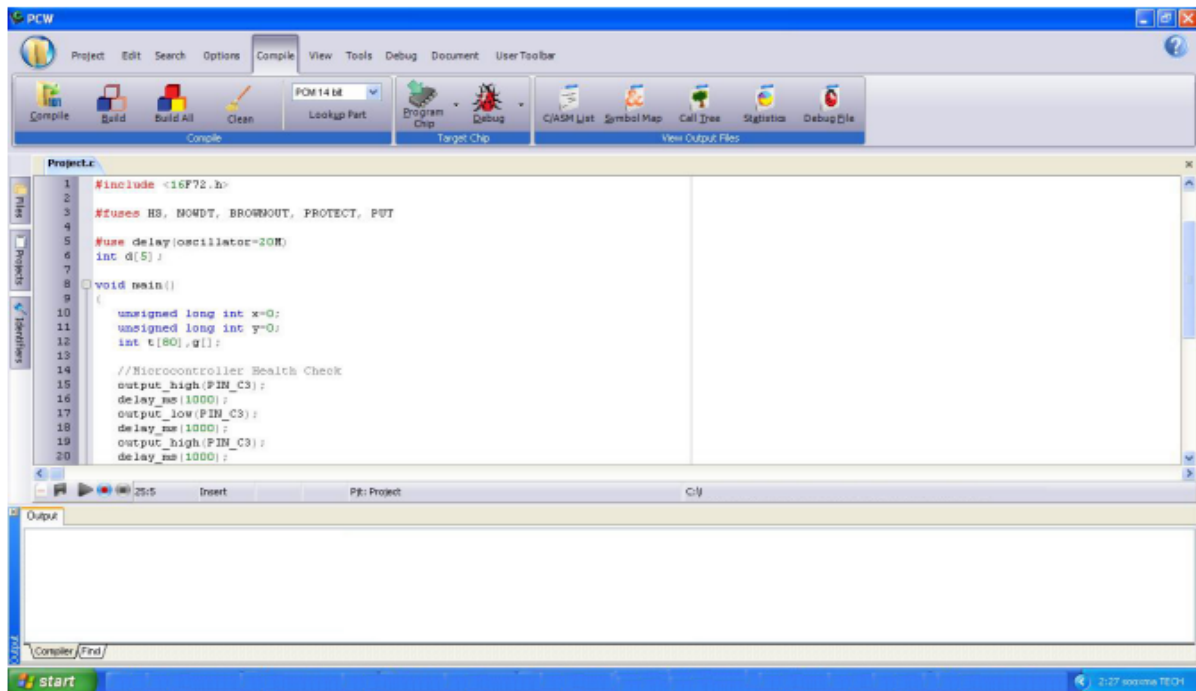


Fig. 5.5: Picture of compiling a new file using PIC C compiler

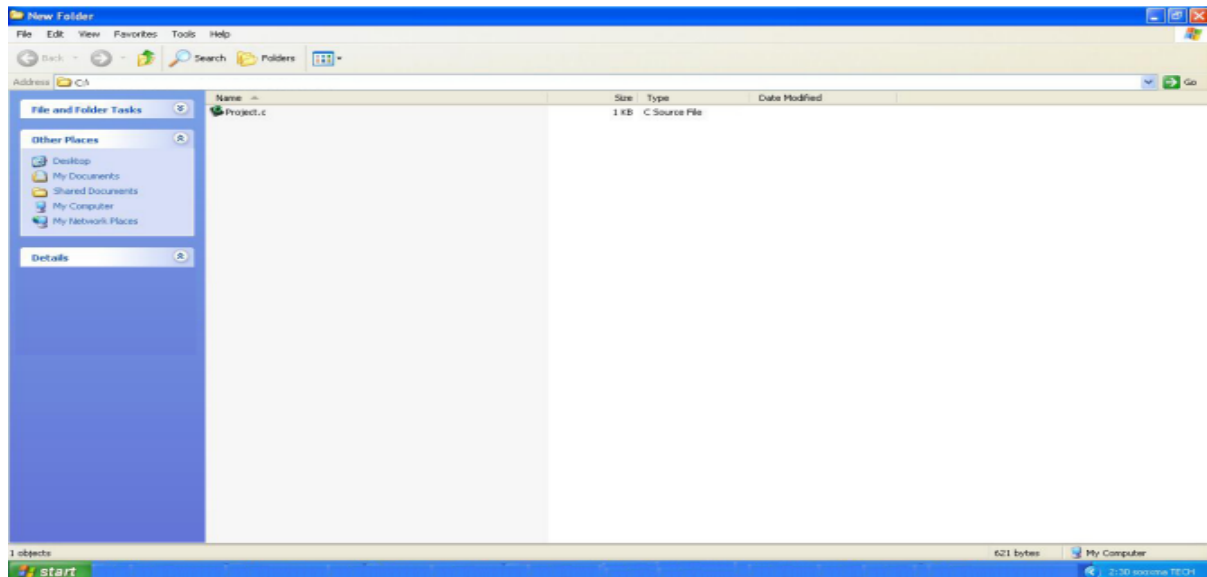


Fig. 5.6: Picture of compiling a project.c file using PIC C compiler

- You can then start to write the source code in the window titled 'project.c' then before testing your source code; you have to compile your source code, and correct eventual syntax errors.

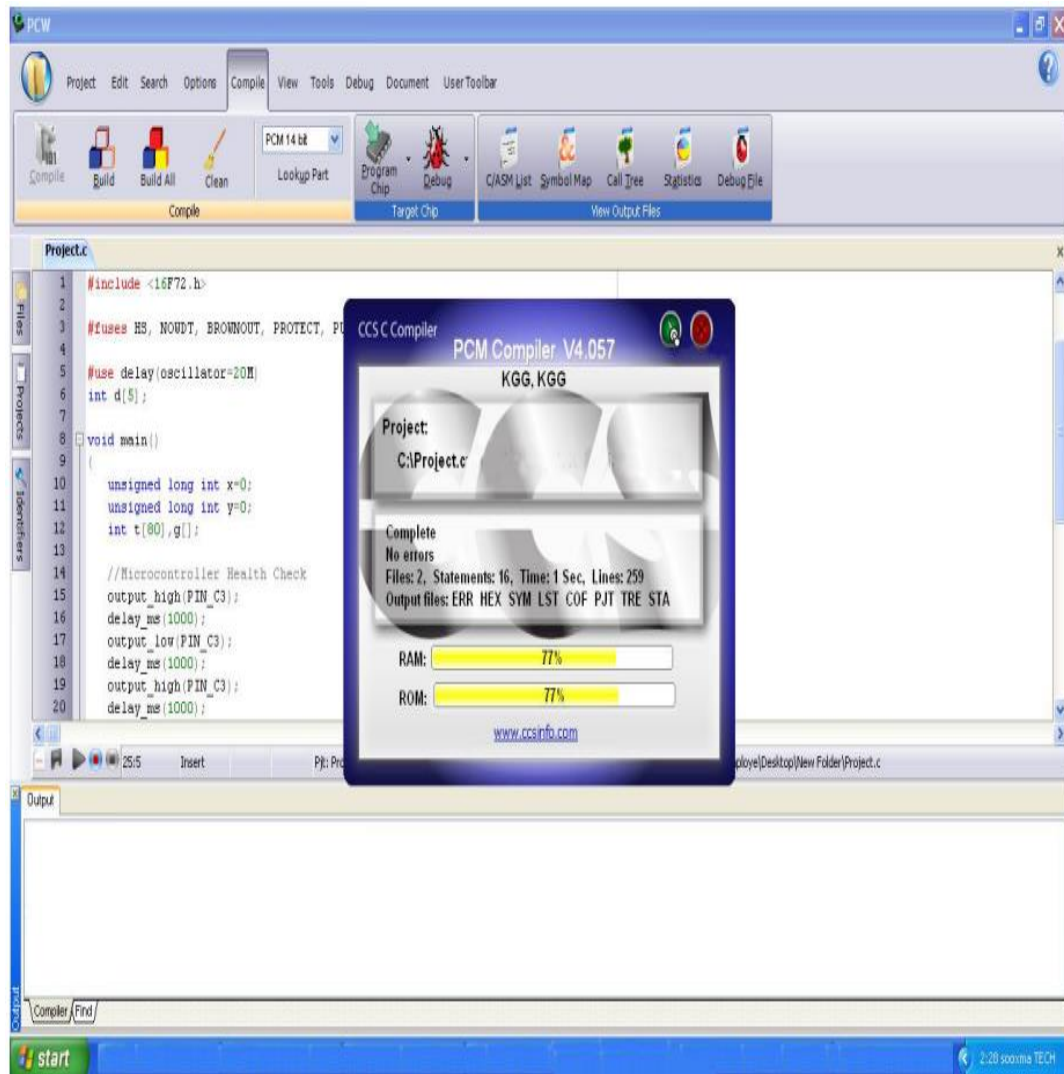


Fig. 5.7: Picture of checking errors and warnings using PIC C compiler

- By clicking on the compile option, the .hex file is generated automatically.
- This is how we compile a program for checking errors and hence the compiled program is saved in the file where we initiated the program.

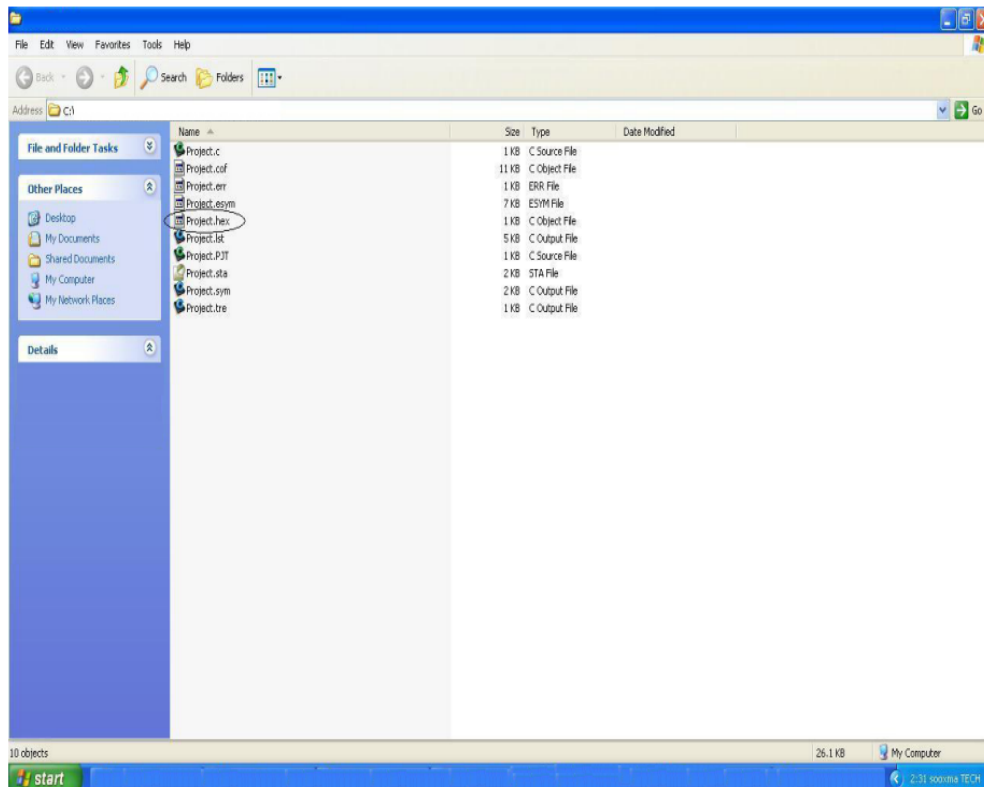


Fig. 5.8: Picture of .hex file existing using PIC C compiler

After compilation, the next step is simulation. Here the first circuit is designed in Express PCB using Proteus 7 software and then simulation takes place followed by dumping. The simulation steps are as follows:

- Open Proteus 7 and click on IS1S6.
- Now it displays a PCB where the circuit is designed using a microcontroller. To design circuit components are required. So click on the component option.
 1. Now click on the letter 'p', then under that select PIC16F73 ,other components related to the project and click OK.
 2. The PIC 16F73 will be called your “Target device”, which is the final destination of your source code.

5.7 Dumping Steps

The steps involved in dumping the program edited in proteus 7 to microcontroller are shown below:

Initially before connecting the program dumper to the microcontroller kit the window appeared as shown below.

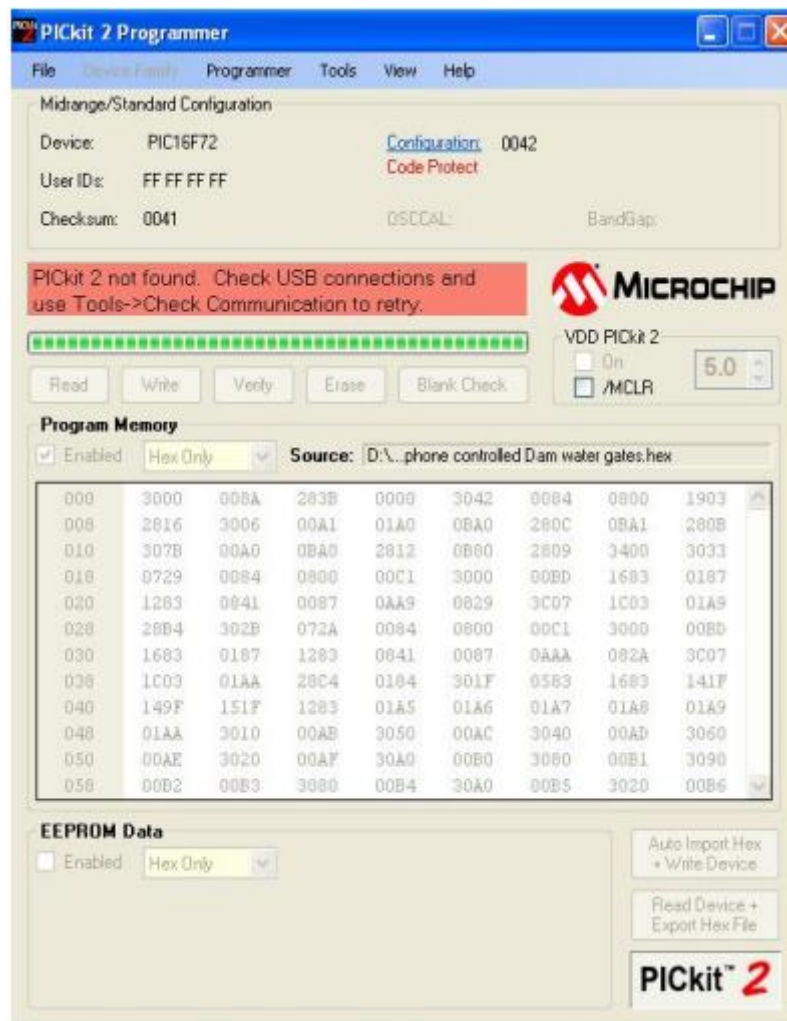


Fig. 5.9: Picture of program dumper window

- Select Tools option and click on Check Communication for establishing a connection as shown in below window.

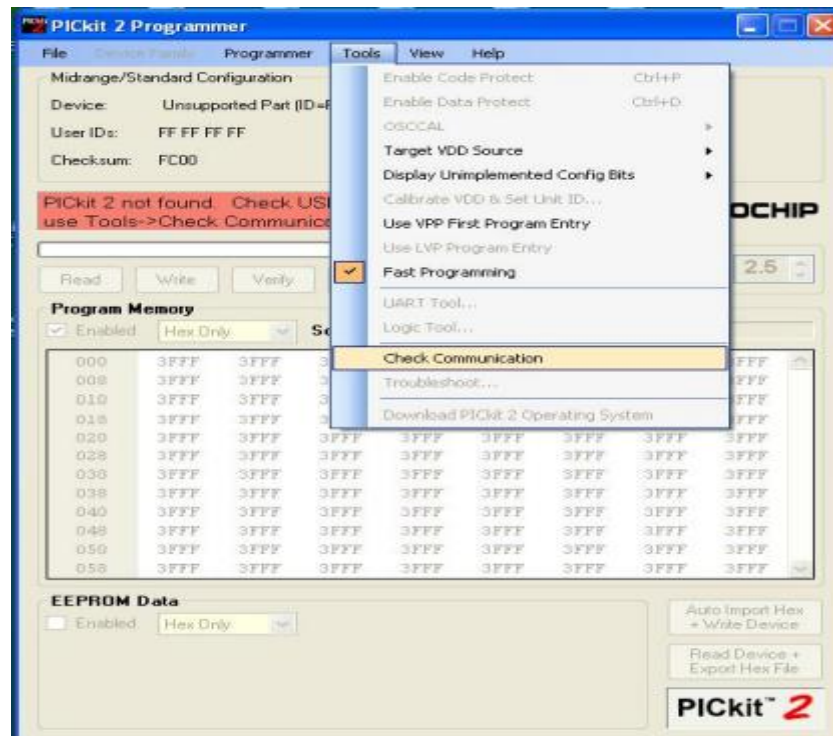


Fig. 5.10: Picture of checking communications before dumping program into microcontroller

- After connecting the dumper properly to the microcontroller kit the window appears as shown below.

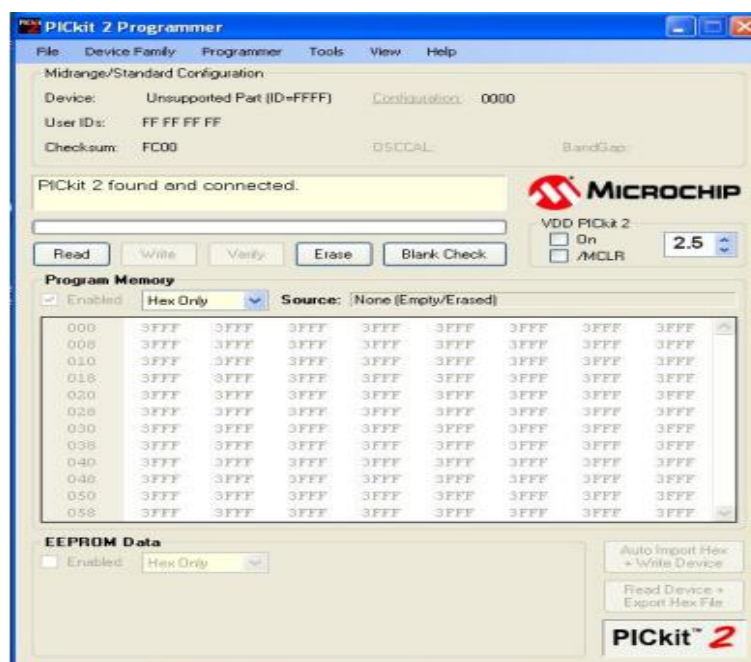


Fig. 5.11: Picture after connecting the dumper to microcontroller

- Again by selecting the Tools option and clicking on Check Communication the microcontroller gets recognized by the dumper and hence the window is as shown below.

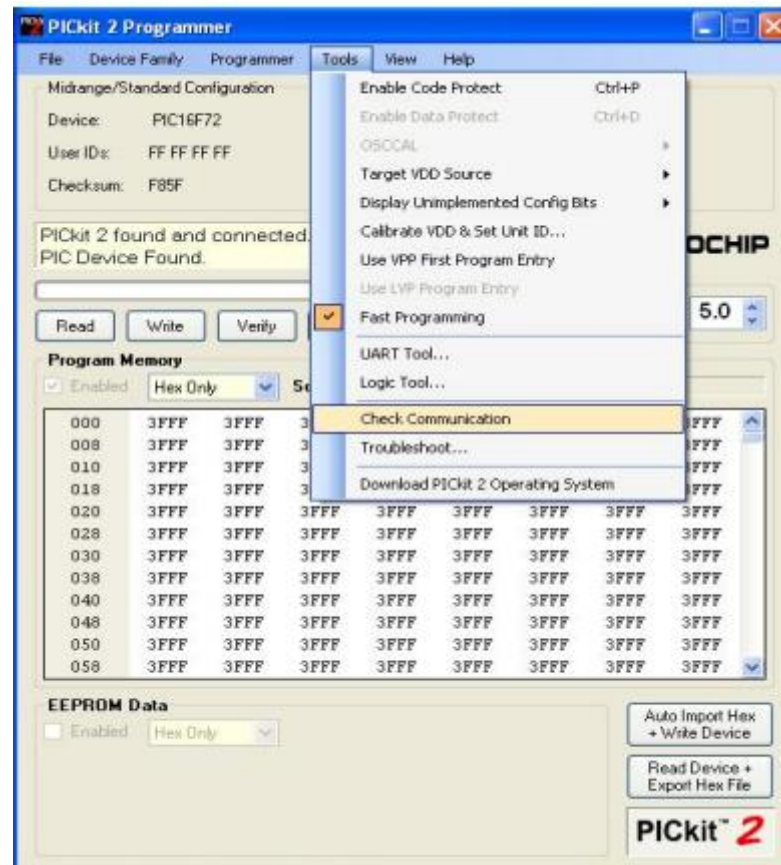


Fig. 5.12: Picture of dumper recognition to microcontroller

- Import the program which is a '.hex' file from the saved location by selecting File option and clicking on 'Import Hex' as shown in below window.

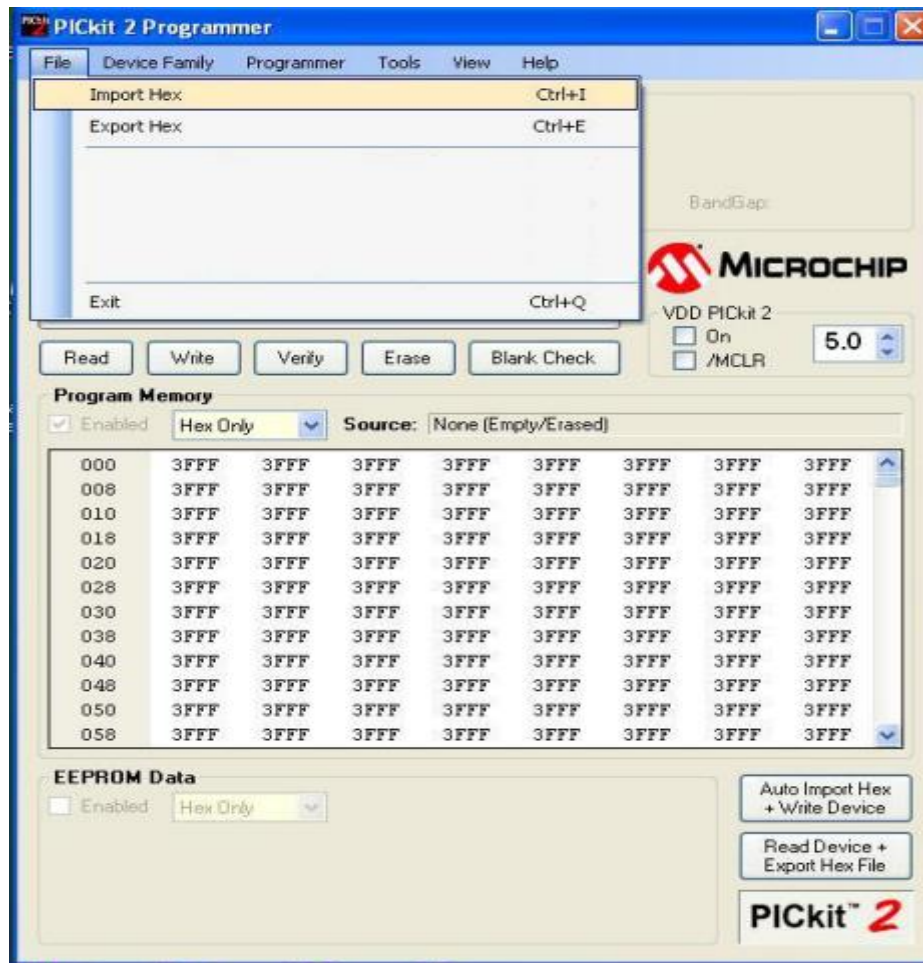


Fig. 5.13: Picture of program importing into the microcontroller

- After clicking on the 'Import Hex' option we need to browse the location of our program and click the 'prog.hex' and click on 'open' for dumping the program into the microcontroller.
- Successful dumping of the program window is as shown below.

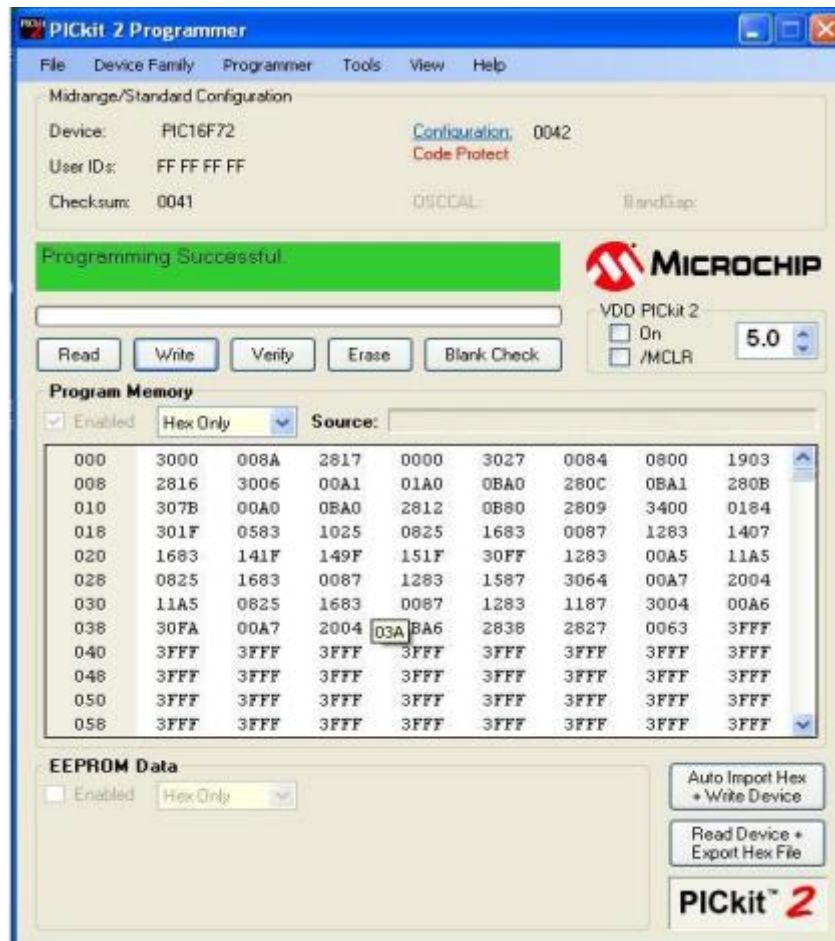


Fig. 5.14: Picture after program dumped into the microcontroller

5.8 Program

```
#include <LiquidCrystal.h>
```

```
#include <stdio.h>
```

```
LiquidCrystallcd(6, 7, 5, 4, 3, 2);
```

```
unsigned char rcv,count,gchr,gchr1;
```

```
char pastnumber[11];
```

```
intsti=0;
```

```
String inputString = ""; // a string to hold incoming data
```

```
booleanstringComplete = false; // whether the string is complete
```

```
int relay1    = 8;
int relay2    = 9;
int relay3    = 10;
```

```
int buzzer    = 13;
```

```
void okcheck()
{
    unsigned char rcr;
    do{
        rcr = Serial.read();
    }while(rcr != 'K');
}
```

```
void beep()
{
    digitalWrite(buzzer, LOW);delay(2500);digitalWrite(buzzer, HIGH);
}
```

```
void setup()
{
    Serial.begin(9600);serialEvent();
}
```

```
pinMode(relay1, OUTPUT);
pinMode(relay2, OUTPUT);
pinMode(relay3, OUTPUT);
pinMode(buzzer, OUTPUT);
```

```
digitalWrite(relay1,LOW);
digitalWrite(relay2,LOW);
digitalWrite(relay3,LOW);
digitalWrite(buzzer,HIGH);
```

```

lcd.begin(16, 2);lcd.cursor();
lcd.print("Substation Monitoring");
delay(2000);

gsminit();

lcd.clear();

digitalWrite(relay1,HIGH);
digitalWrite(relay2,HIGH);
digitalWrite(relay3,HIGH);

serialEvent();
}

float current1=0,current2=0,current3=0;
float voltage1=0,voltage2=0,voltage3=0;
float freq1=0,freq2=0,freq3=0;
int a=0,b=0,c=0;
float tempc=0;
float tempf=0;

void loop()
{

lcd.begin(16, 2);lcd.cursor();
    current1 = analogRead(A1);
    current1 = (current1 * 0.23);
lcd.setCursor(0,0);lcd.print("C1:");convertl(current1);
delay(100);

    current2 = analogRead(A2);
    current2 = (current2 * 0.23);

```

```
lcd.setCursor(8,0);lcd.print("C2:");convertl(current2);  
delay(100);
```

```
    current3 = analogRead(A3);  
    current3 = (current3 * 0.23);  
lcd.setCursor(0,1);lcd.print("C3:");convertl(current3);  
delay(100);
```

```
/*
```

```
tempf = analogRead(A0);  
tempf = ((tempf * 5000) / 1024);  
tempf = (tempf/10);  
tempc = ((tempf - 32.0) * (5.0/9.0));  
*/
```

```
tempc = analogRead(A0);  
tempc = (tempc*0.48828125);  
lcd.setCursor(8,1);lcd.print("T:");convertl(tempc);  
delay(100);
```

```
if(tempc> 45)  
    {beep();  
    delay(4000); delay(5000);  
    Serial.write("AT+CMGS=\"");  
    Serial.write(pastnumber);  
    Serial.write("\r\n"); delay(3000);  
    Serial.write("High Temp:");converts(tempc);  
    Serial.write(0x1A);  delay(4000); delay(5000);  
    }
```

```
delay(3000);  
lcd.begin(16, 2);lcd.cursor();  
if(current1 <= 1){voltage1=0;freq1=0;}  
    else{voltage1=230;freq1=55;}
```

```
if(current2 <= 1){voltage2=0;freq2=0;}
```

```

    else{voltage2=230;freq2=55;}

if(current3 <= 1){voltage3=0;freq3=0;}
    else{voltage3=230;freq3=55;}

lcd.setCursor(0,0);lcd.print("V1:");convertl(voltage1);
lcd.setCursor(8,0);lcd.print("V2:");convertl(voltage2);
lcd.setCursor(0,1);lcd.print("V3:");convertl(voltage3);
delay(3000);

lcd.begin(16, 2);lcd.cursor();
lcd.setCursor(0,0);lcd.print("F1:");convertl(freq1);
lcd.setCursor(8,0);lcd.print("F2:");convertl(freq2);
lcd.setCursor(0,1);lcd.print("F3:");convertl(freq3);
delay(3000);

if(current1 > 100 && a==0)
    {beep();
        a=1;          delay(4000); delay(5000);
Serial.write("AT+CMGS=\");
Serial.write(pastnumber);
Serial.write("\r\n"); delay(3000);
Serial.write("High Current Over Load1:");converts(current1);
Serial.write(" Temp:");converts(tempc);
Serial.write(" Voltage1:");converts(voltage1);
Serial.write(" Frequency1:");converts(freq1);
Serial.write(0x1A);  delay(4000); delay(5000);
    }
if(current2 > 100 && b==0)
    {beep();
        b=1;          delay(4000); delay(5000);
Serial.write("AT+CMGS=\");
Serial.write(pastnumber);

```

```

Serial.write("\r\n"); delay(3000);
Serial.write("High Current Over Load2:");converts(current2);
Serial.write(" Temp:");converts(tempc);
Serial.write(" Voltage2:");converts(voltage2);
Serial.write(" Frequency2:");converts(freq2);
Serial.write(0x1A);  delay(4000); delay(5000);
    }
if(current3 > 100 && c==0)
    {beep();
        c=1;          delay(4000); delay(5000);
Serial.write("AT+CMGS=\");
Serial.write(pastnumber);
Serial.write("\r\n"); delay(3000);
Serial.write("High Current Over Load3:");converts(current3);
Serial.write(" Temp:");converts(tempc);
Serial.write(" Voltage3:");converts(voltage3);
Serial.write(" Frequency3:");converts(freq3);
Serial.write(0x1A);  delay(4000); delay(5000);
    }

```

```

if(stringComplete)
{
    if(inputString[1] == '1')
    {
digitalWrite(relay1,HIGH);
    }
    if(inputString[1] == '2')
    {
digitalWrite(relay1,LOW);
    }

    if(inputString[1] == '3')
    {

```

```

digitalWrite(relay2,HIGH);
    }
    if(inputString[1] == '4')
    {
digitalWrite(relay2,LOW);
    }

    if(inputString[1] == '5')
    {
digitalWrite(relay3,HIGH);
    }
    if(inputString[1] == '6')
    {
digitalWrite(relay3,LOW);
    }
inputString = "";
stringComplete = false;
    }
}

void serialEvent()
{
while(Serial.available())
    {
        char inChar = (char)Serial.read();
if(inChar == '*')
        {sti=1;
        }
if(sti == 1)
        {
inputString += inChar;
        }
if(inChar == '#')
        {sti=0;

```

```
stringComplete = true;
```

```
    }
```

```
    }
```

```
}
```

```
intreadSerial(char result[])
```

```
{
```

```
    inti = 0;
```

```
    while (1)
```

```
    {
```

```
        while (Serial.available() > 0)
```

```
        {
```

```
            char inChar = Serial.read();
```

```
            if (inChar == '\n')
```

```
            {
```

```
                result[i] = '\0';
```

```
Serial.flush();
```

```
        return 0;
```

```
    }
```

```
    if (inChar != '\r')
```

```
    {
```

```
        result[i] = inChar;
```

```
    i++;
```

```
    }
```

```
}
```

```
}
```

```
}
```

```
void gsmininit()
```

```
{
```

```
Serial.write("AT\r\n"); okcheck();
```

```
Serial.write("ATE0\r\n"); okcheck();
```

```
Serial.write("AT+CMGF=1\r\n"); okcheck();
```

```
Serial.write("AT+CNMI=1,2,0,0\r\n"); okcheck();
```

```
Serial.write("AT+CSMP=17,167,0,0\r\n"); okcheck();
```

```
lcd.clear();
```

```
lcd.print("SEND MSG STORE");
```

```
lcd.setCursor(0,1);
```

```
lcd.print("MOBILE NUMBER");
```

```
do{
```

```
rcv = Serial.read();
```

```
}while(rcv != '*');
```

```
readSerial(pastnumber);
```

```
lcd.clear();
```

```
lcd.print(pastnumber);
```

```
pastnumber[10]='\0';
```

```
delay(4000); delay(4000);
```

```
Serial.write("AT+CMGS=\"");
```

```
Serial.write(pastnumber);
```

```
Serial.write("\r\n"); delay(3000);
```

```
Serial.write("Mobile no. registered\r\n");
```

```
Serial.write(0x1A); delay(4000);delay(4000);
```

```
}
```

6. TECHNICAL ARCHITECTURE

6.1 Block Diagram

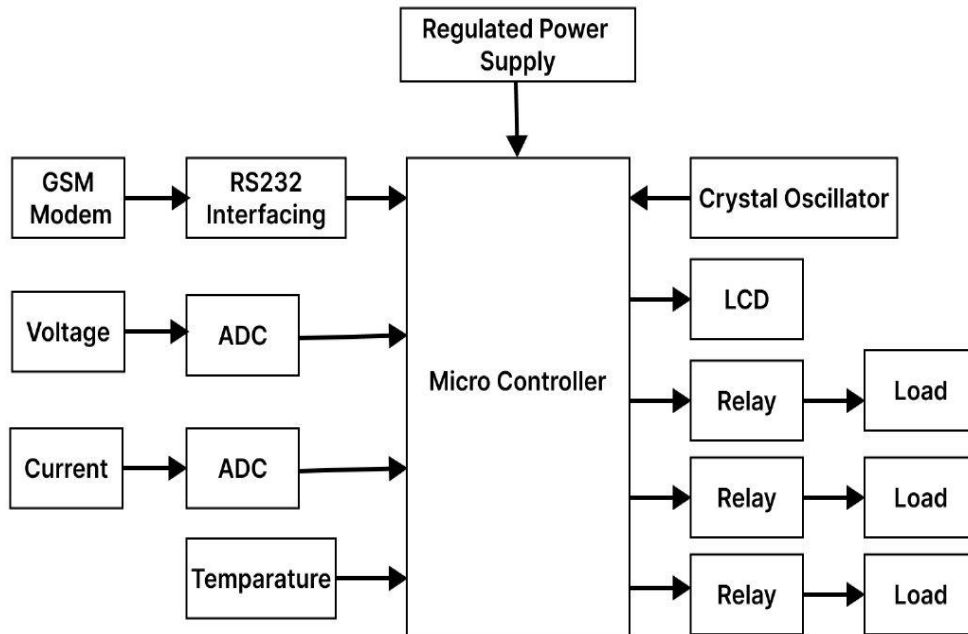


Fig. 6.1: Block diagram of substation monitoring and controlling based on microcontroller through GSM modem.

This system acquires a remote substation's parameters like voltage, current, frequency, etc., via GSM network using GSM modem or mobile. It allows the remote operator to operate relays, circuit breakers and buzzers, and facilitates the display of various parameters in the LCD that is interfaced to the microcontroller.

A microcontroller acts as a central controlling device that accepts the inputs and correspondingly controls the outputs, as shown in the above figure. The microcontroller is programmed in such a way that it sends the input parameters in the field side to a remote GSM mobile through the GSM networks, periodically.

Similarly, it receives the control signals that are sent from the operators, and in accordance with those signals, it controls the relays and buzzers. At times when current exceeds their limits,

the microcontroller automatically trips the circuit breakers and sends that information to the user by SMS.

This system also can automatically send the real time electrical parameters periodically (based on time settings) in the form of SMS. This system can be designed to send SMS alerts whenever the Circuit Breaker trips or whenever the Voltage or Current exceeds the predefined limits. This project makes use of an onboard computer which is commonly termed as microcontroller.

This onboard computer can efficiently communicate with the different sensors being used. The controller is provided with some internal memory to hold the code. This memory is used to dump some set of assembly instructions into the controller. And the functioning of the controller is dependent on these assembly instructions. The controller is programmed using Embedded C language

Thus, this is a simple and low-cost type of GSM-based project that helps students to get a basic idea about the substation automation. With this system, a user can add more than two mobile numbers to receive and control the substation equipment.

6.2 Schematic Diagram

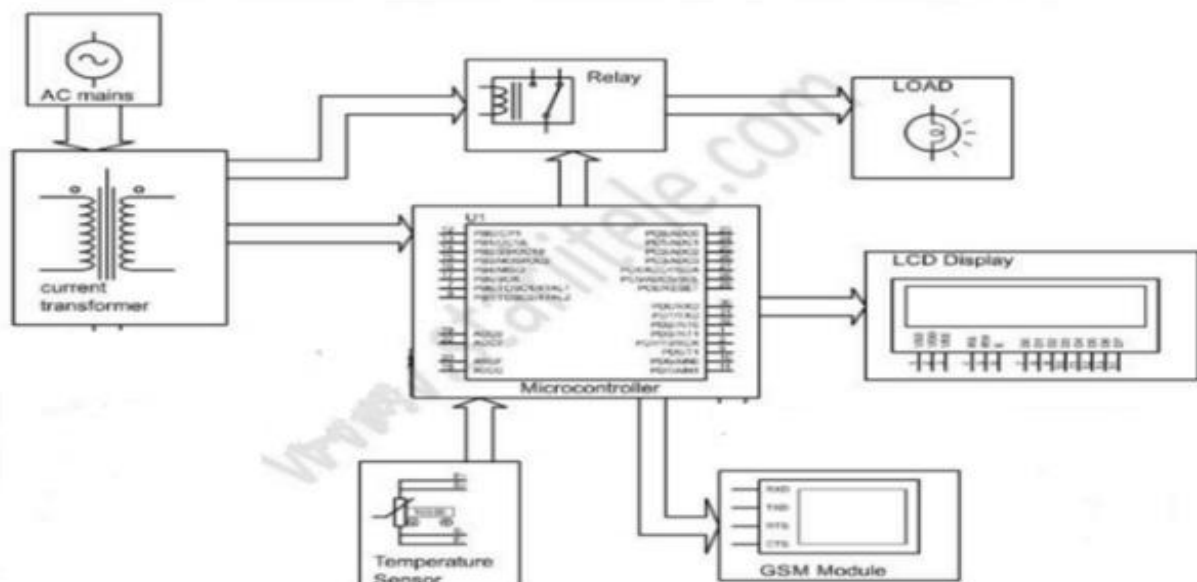


Fig. 6.2: Schematic diagram of Substation Monitoring and Controlling Based on Microcontroller Using GSM Modem

Here in the proposed prototype we have used Arduino Uno as our primary microcontroller. It will work as the heart of the system; all other measurement circuitries will be interfaced through this. All the detailed specifications will be discussed in the next chapter, nevertheless let me take you through the basic components for better understanding the rest of the operation. Besides the microcontroller we have used current sensor, voltage sensor, temperature sensor, frequency measurement unit, buzzer and relay; and to demonstrate the load we have used a fan and a bulb.

Alongside we also have used a supply unit, consisting of a transformer, which converts 230 Volt AC to 12 Volt AC then it is passed through bridge rectifier unit which converts this 12 Volt AC to 12 Volt DC which is pulsating in nature which is then fed to the capacitor which work as a filter, makes the pulsating DC to smooth DC. As a lot of our components like Arduino Uno and some of the sensors as well require 5 Volt regulated DC, that is why this 12 Volt DC is fed to 7805 Voltage regulator which makes it to 5 Volt regulated DC.

In case relay need high amount of current for operation, we have to make some arrangements for that. For that amplification arrangement in case of relay we have used a relay driver and for buzzer we have used two BC 547 transistor in Darlington pair configuration. For operation of the green and when we give supply to our prototype the display shows a welcome message and simultaneously all the sensor start sensing the current, voltage, frequency and temperature and update all the real time values to the server as well as shows on the display.

It compares all the real time values with the predefined values, if any of the values exceed predefined values it sends a fault alert to the relay as well as update it on the display. If the fault exists for the pre-set time then relay isolates the loads from the rest of the system. In the meantime, comparison goes on as before, if the fault gets cleared relays reconnect the loads with the rest of the system

6.3 Advantages

1. Less Copper

Digital Substations utilize less copper by (1) replacing copper cables between switchyard and relay house by fiber optics, (2) replacing horizontal wiring between protection and control IEDs with IEC 61850, (3) reducing the number of connections between primary apparatus and redundant process interface modules and (4) reduction in cables for power supply and other connection for primary equipment.

2. Less Transport

In digital substation, more than 30 tons of material can be saved [45]. This helps in less transportation. The fiber optic installation reduces the cabling under 90% and by using optical instead of conventional CTs achieves an 80% weight reduction on CTs.

3. Space Reduction

IED installations require less space than conventional IOs. A space reduction around 30% to 60% for panels is possible. This results in higher integration of control and protection systems and reduction of switchyard footprint.

4. Less Installation and Outage Time

Digital substations cause faster installation of control systems due to reduction in number of panels for installation. There are fewer cables to be pulled, connected and tested. This causes reduction of feeder outage time by 40 to 50% during secondary system upgrades. This helps in full system test from process IO to protection, control and SCADA system offsite and Installation of new FO based system while the station is in service

5. Operational Cost Reduction

The Supervision of all exchanged data reduces the need for periodic maintenance testing and permanent supervision enables fast and precise actions in case of failures. IEC 61850 testing and simulation features enable fast and save isolation and testing of protection functions. Standard compliance enables efficient future retrofits of the secondary system.

6.4 Disadvantages

1. Continuous communication network should be available.
2. During emergency situation, network congestion or performance are big challenges in smart grid system.
3. Cellular network providers do not provide guaranteed service in abnormal situations such as wind storm, heavy rain and lightning conditions.
4. Some smart meters can be hacked which can be used to increase or decrease the demand for power.
5. It is expensive to install smart meter compare to traditional old electricity meter.

6.5 Applications

This can be practically implemented in real time

1. Transporting energy.
2. Transforming voltage.
3. Distributing energy.
4. Controlling power flow.
5. Ethernet-based communication on IEC 61850: Digitalization of station level.
6. Digitalization of process level: Introduction of process bus and NCITs.
7. Asset management support: Use of data from the digital substation for asset management.
8. Integrated engineering: Consistent, efficient engineering.
9. Network control support: Use of data from the digital substation supports network control.

6.6 Testing and Result

Case 1: First we have to insert SIM, switch on the power supply and then check signal after getting signal press reset button and register the mobile number and start the process.



Fig. 6.3: LCD displaying to send the message to the modem

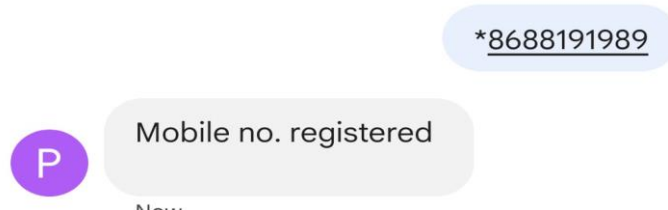


Fig. 6.4: Mobile number is registered in the modem

Case 2: Creating one over current condition current ratings are $C1 = 67\text{mA}$ $C3 = 160\text{mA}$. Third bulb is under over current condition. It is turned off by sending a command.

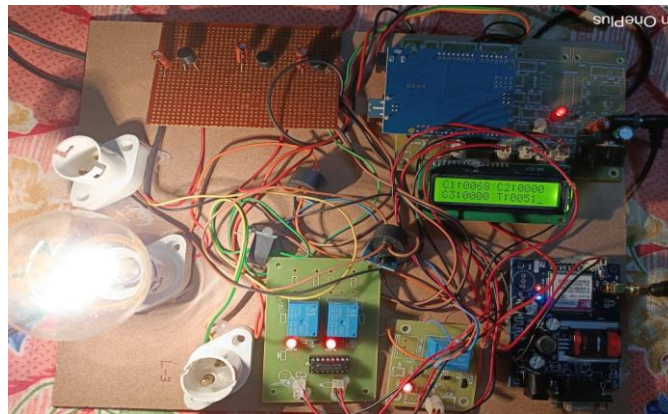


Fig. 6.5: One bulb is given as the load

Case 3: Creating one over current condition current ratings are $C1 = 67\text{mA}$ $C3 = 160\text{mA}$. Third bulb is under over current condition. It is turned off by sending a command.



Fig. 6.6: Two bulbs are given as the load

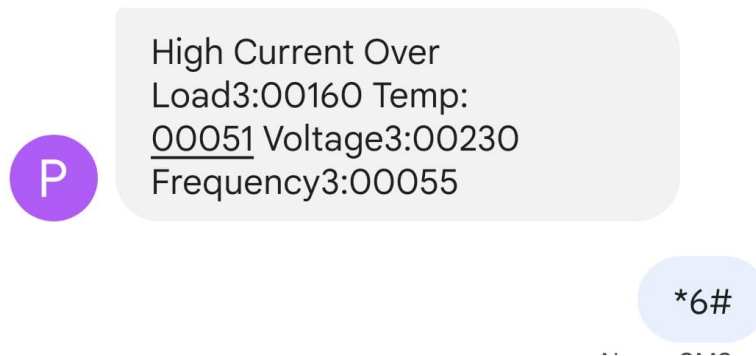


Fig. 6.7: Message from the Modem and giving Command

Case 4: We are connecting three bulbs, in this case two bulbs shows over current condition here we can off all either three bulbs or two bulbs.



Fig. 6.8: Three bulbs are placed simultaneously

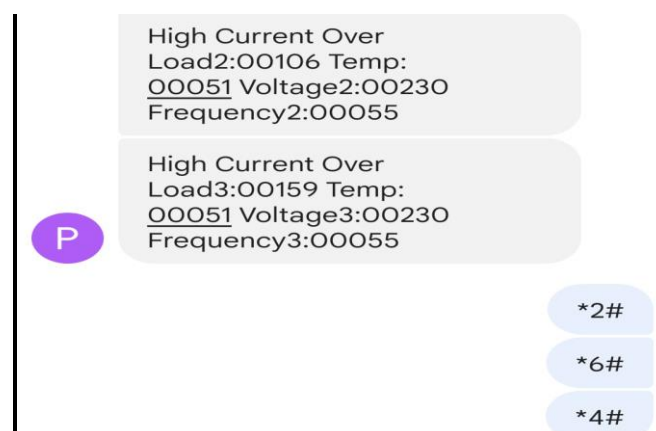


Fig. 6.9: Messages from modem showing overload and sending commands to turn off the loads.

The project “Substation Monitoring and Controlling Based on Microcontroller by Using GSM Modem” This project is also planned by running an Electromagnetic Relay to secure the electrical circuits. Whenever the electrical parameters exceed the predefined values, this Relay is triggered. Thus, it senses the fault and gives the signal to registered mobile. By receiving commands elimination of fault can be done when the system overloads.

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

Integrating features of all the hardware components used have been developed in it. Presence of every module has been reasoned out and placed carefully, thus contributing to the best working of the unit. It can improve the quality of power transferred and provide uninterrupted power supply. Also real time monitoring of different parameters is done which can provide safety to the substation and its equipment. Secondly, using highly advanced IC's with the help of growing technology, the project has been successfully implemented. Thus the project has been successfully designed and tested.

7.2 Future Scope

The main drawback of the system is the Bluetooth wireless communication supports only for limited distance. This can be eliminated using GSM technology which can increase the distance using SMS technology.

The world has gone digital and that includes the electricity industry. Providing electricity has always been challenging, but new technologies increased the complexity of that challenge when moving from the 20th century to the 21st century. The industry had to bridge the gap between aging analog devices and digital technologies.

The future of digital substations leads to expect that there will be more technologies and developments, for example centralized protection, that allow substations, or rather, digital substations to evolve to make them even more efficient and safe.

By connecting temperature, gas, sensors to the system we can get the temperature of dangerous zones in through message itself instead of sending human to there and facing problems at field. The project can be extended by using relay which can be operated automatically depending upon the inputs of sensors.

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